Gender and CGPA across batches

iGEM IISER Kolkata 2026 · Inclusivity ·

What this is

The Dean of Academic Affairs (DoAA, Dr. Rangeet Bhattacharyya) very kindly provided us with the anonymised CGPA data for men and women. We use two sets: **mean CGPA by gender for each semester (1 to 10) for eight batches**, and **mean GPA by gender in seven subject areas for six batches, covering the first four semesters**. Batches are renamed Batch 1 to Batch 8 in order of admission year. The first two years (semesters 1 to 4) are before the major is chosen (semester 5).

How the numbers were obtained. The semester data (obtained as eight line graphs), were obtained by measuring the lines pixel by pixel, which gives values accurate to about ±0.02 CGPA points. The subject data was received as a table.

Findings

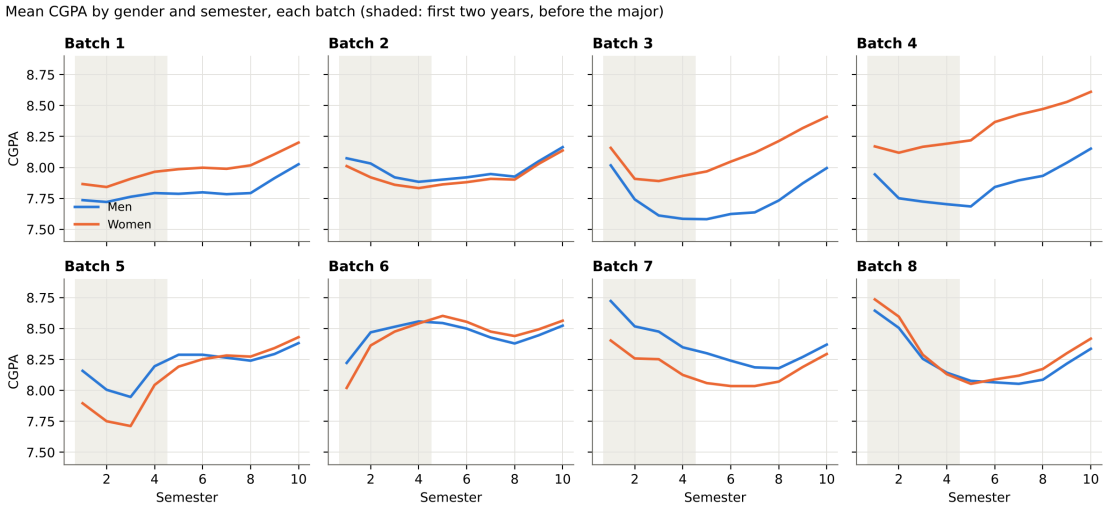
1. **In general, women seem to tend to do better in academics grade-wise.** The general trend (in the data set that we observed) shows that, with some exceptions, women generally to better in academics (or, atleast, are mostly at par with their male counterparts). Therefore, the low number of women students at IISER Kolkata cannot be simply attributed to "capability".
2. **In the first two years there is no consistent direction.** At semester 4, women's mean CGPA is higher in 3 of the 8 batches and lower in 5. The gap ranges from −0.22 to +0.49 points. The average over the eight batches (+0.07 at semester 4) is pulled up by Batches 3 and 4, where women lead by 0.35 and 0.49. Without those two it is negative (−0.08 averaged over semesters 1 to 4).
3. **Later, women are level or ahead in most batches.** At semester 10, women's mean is higher in 6 of the 8 batches (range −0.08 to +0.46), and the eight-batch average is +0.14. Without Batches 3 and 4 it is +0.04.
4. **Batch-to-batch differences are larger than the gender gap.** Batch 4 has a gap above 0.4 for most of the course and Batch 7 a gap near −0.2 in the first years. The gender gap is smaller than the variation between batches except in a few batches.
5. **Gaps in the early semesters shrink in the batches where men led.** In Batches 5 and 6 the early male lead (−0.23 and −0.09 averaged over semesters 1 to 4) closes by semester 5 to 7. In Batch 7 it narrows from −0.31 to −0.08.
6. **By subject, the pattern differs, and varies by batch.** Women's mean GPA is higher in humanities (in all 6 batches, on average +0.16), biology (+0.25) and geology (+0.24), and lower in physics (−0.18), mathematics (−0.15) and computer science (−0.12). In Batches 3 and 4 women lead in nearly every

subject; in Batches 5 to 7 they trail in physics, mathematics and computer science. Chemistry is split (average +0.08). We explore this in the next report in detail.

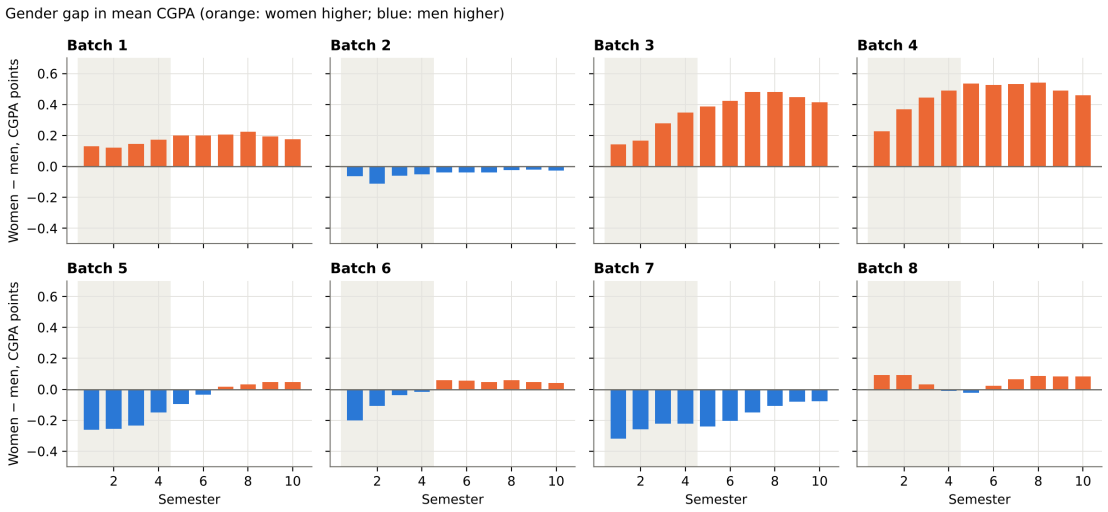
What this does and does not tell us

- **Marks do not point to women being weaker.** If anything the later gap is small and slightly in favour of women. ***The shortfall in women in science at IISERs, which our data report shows, is therefore not explained by grades among those who are here.*** Where it comes from (who applies, who is admitted, who chooses which major, who continues to a PhD) these numbers cannot say.
- **Selection can produce this pattern without any real advantage.** If women face higher barriers to getting in or staying, the women who are present may be a more selected group. The same logic applies to the subject differences, where the number of women in each major may be small.
- **CGPA is cumulative.** The line for semester 6 includes semesters 1 to 5, so a change between semesters is smoothed.
- **The subject numbers cover the first four semesters only.** In semesters 1 and 2 every student studies every subject; in semesters 3 and 4 each student studies three subjects of her or his choice, so later averages already reflect choices. The values are GPA in courses of each subject area, which is why they are lower than the CGPA in the graphs (for example, Batch 3 men average about 7.5 across subjects against about 8.0 in the CGPA graph). They cover Batches 3 to 8 only.
- **Spread.** Without the number of women and men in each batch (and the standard deviation of CGPA) we cannot give confidence intervals or tests. Women are fewer in each batch, so their means are less stable. A gap of 0.1 to 0.2 points could easily be chance in a small group. We will look into it in the later phases if possible.
- **Uncertainty.** Given that the number of women students at IISER Kolkata is already low (around 27 percent on an average), even small differences can give rise to major fluctuations. This should be kept in mind before drawing any generalized conclusions.

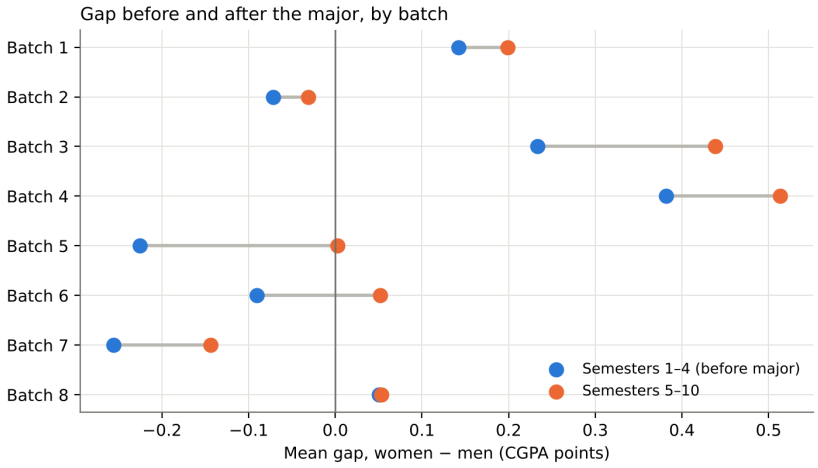
Graphs



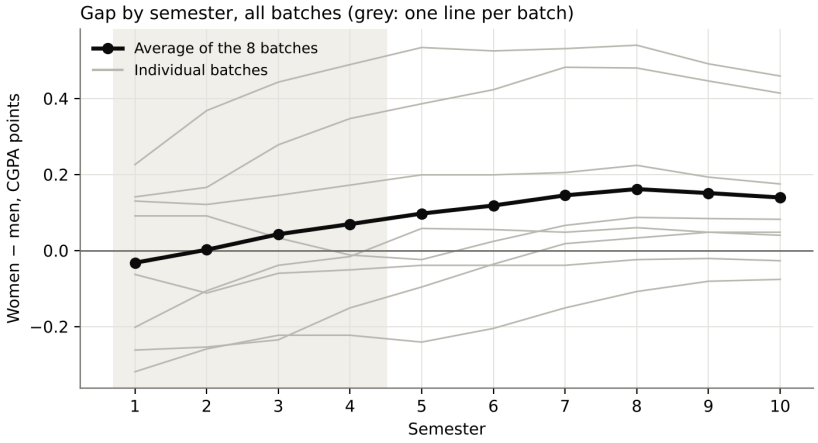
Mean CGPA by gender and semester for each batch. Shaded: first two years, before the major.



Gap (women minus men) by semester for each batch. Clearly, the data tends to show better average performance amongst the women



Average gap in the first two years and in later semesters, by batch.



Gap by semester, all batches together. Grey lines are individual batches.

The next steps

1. **Look at the subject-wise GPA data for women and men in more detail.** The subject numbers (first four semesters, before the major is chosen) will be analysed separately, subject by subject and batch by batch, to see where women and men differ and whether those differences might relate to who chooses mathematics, physics or another major. This is set out in a separate report, *Subject-wise GPA by gender*.
2. **The number of women and men in each batch** (and in each subject, if possible). This is the most important gap, because it decides which differences can be trusted.
3. **Standard deviation or the quartiles of CGPA** by gender, so we can express gaps in a way that means something.
4. **SGPA by semester,** which would show single-semester performance.
5. **Which major each student chose, by gender,** so we can see whether the first-two-year results relate to who chooses which major.
6. **How many students left, changed programme or took a break,** by gender and batch.
7. **The same subject-wise data for Batches 1 and 2, and for semesters 1 and 2 separately** (when everyone takes every subject), which removes the effect of students choosing subjects.